

Buffon's Needle

Randomness, Geometry, and an Estimate of π

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Outline

- 1 Why This Problem
- 2 Geometry to Formula
- 3 Simulation
- 4 Extensions and Uses
- 5 Demo

A Simple Puzzle with a Big Idea

- Buffon posed the needle problem in the 18th century.
- A random experiment can reveal a precise constant: π .
- It is an early example of geometric probability.
- It also anticipates the Monte Carlo idea: estimate by repeated random trials.

Core question

If we drop a needle onto equally spaced parallel lines, how often does it cross a line?

Setup and Crossing Rule

Experiment setup

- Line spacing: d
- Needle length: $l \leq d$
- Angle to the lines: $\theta \in [0, \pi]$
- Midpoint distance to the nearest line: $x \in [0, d/2]$

Crossing condition

A crossing happens when

$$x \leq \frac{l}{2} \sin \theta.$$

Special case used in our demo

We set $l = d = 1$, so only x and θ need to be sampled.

Theory: Sample Space and Favourable Region

By symmetry

Instead of using every angle in $[0, \pi]$, we can work on the smaller sample space

$$x \in \left[0, \frac{d}{2}\right], \quad \theta \in \left[0, \frac{\pi}{2}\right].$$

The total area is therefore

$$A_{\text{total}} = \frac{d}{2} \cdot \frac{\pi}{2} = \frac{\pi d}{4}.$$

Crossing region

The needle crosses a line when

$$x \leq \frac{l}{2} \sin \theta.$$

So for each fixed θ , the allowed values of x are

$$0 \leq x \leq \frac{l}{2} \sin \theta.$$

Favourable area

$$A_{\text{fav}} = \int_0^{\pi/2} \int_0^{(l/2) \sin \theta} dx \, d\theta.$$

Theory: Why π Appears

Evaluate the integral

$$A_{\text{fav}} = \int_0^{\pi/2} \frac{l}{2} \sin \theta \, d\theta = \frac{l}{2}.$$

Hence

$$P(\text{cross}) = \frac{A_{\text{fav}}}{A_{\text{total}}} = \frac{l/2}{\pi d/4} = \frac{2l}{\pi d}.$$

Special case in our demo

When $l = d = 1$,

$$P(\text{cross}) = \frac{2}{\pi} \approx 0.63662.$$

Estimator

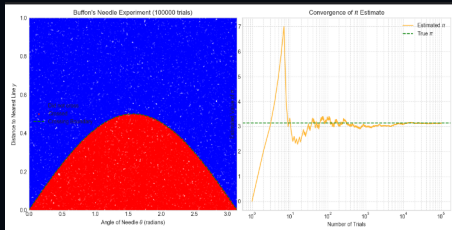
After N throws and C crossings,

$$\hat{P} = \frac{C}{N}, \quad \hat{\pi} \approx \frac{2N}{C}.$$

Simulation Logic and One Sample Run

```
repeat N times:  
  sample theta in [0, pi]  
  sample x in [0, d/2]  
  if x <= (1/2) * sin(theta):  
    C <- C + 1  
  
if l = d = 1:  
  pi_hat <- 2*N / C
```

N	C	$\hat{\pi}$
100	60	3.3333
1,000	628	3.1847
10,000	6,401	3.1245
100,000	63,753	3.1371



Take-away

The estimate fluctuates at first, then settles near π as the number of trials grows.

Beyond a Straight Needle – Moser's Curved Needle

Q

What happens if we replace the straight needle with a curve?

A

For a curved needle, the expected number of intersections depends on the total length rather than the exact shape.

Link

A sufficiently regular curve can be approximated by many tiny straight segments.

Main idea

Bending changes the geometry locally, but the expectation is governed by total length.

Applications in Higher Dimensions

Transformation

From 2D geometric probability to high-dimensional spaces.

Monte Carlo methods

- Estimate high-dimensional volumes, such as hyperspheres.
- Used in finance for risk modelling and probabilistic evaluation.

Random direction simulation

- Generate random directions in high-dimensional space.
- Applications include light scattering in physics and feature optimisation in machine learning.

Take-away

The same random-sampling logic scales far beyond Buffon's original experiment.

Live Web Demo

Open the demo

<https://buffons-needle.byron.wang/>

Open-sourced on GitHub under the MIT License. Copyright belongs to our group.

- Drop 1 or 100 needles, or switch to auto-run.
- Compare C/N with the theoretical value $2/\pi$.
- Watch the estimate $\hat{\pi}$ stabilise in real time.

Closing idea

A simple random rule can recover a famous constant, and the web demo shows that convergence live.



Scan to open the demo on
your phone

Thank You